Mandatory deliverable

LAB | AWS Networking & Virtual Private Cloud (VPC) (Guided) Lab

Skills you will learn

Cloud Computing: Compute & Networking

Introduction

In this hands-on lab, you will design and implement a **custom Virtual Private Cloud (VPC)** architecture in AWS, following best practices in networking and security. This will help you understand how to configure networking components, subnets, route tables, and internet connectivity within AWS.

Learning Objectives

By completing this lab, you will be able to:

1. **Create a custom VPC** with designated public and private subnets.
2. **Set up an Internet Gateway (IGW) and a NAT Gateway (NGW)** for external connectivity.
3. **Configure route tables** to manage traffic flow.
4. **Associate subnets with appropriate route tables**.
5. **Verify connectivity** and troubleshoot potential issues.

Lab Instructions

1. Create a Custom VPC

1. Navigate to **AWS Management Console** → **VPC Service**.
2. Click on **Create VPC**.
3. Enter the following details:
   - **VPC Name**: `prod-vpc`
   - **IPv4 CIDR Block**: `10.0.0.0/16`

Ask AI Tutor

4. Click **Create VPC**.

2. Create Public and Private Subnets

1. Navigate to **Subnets** → Click **Create Subnet**.
2. **Public Subnet:**
   - **Name**: `public-subnet`
   - **VPC**: `prod-vpc`
   - **IPv4 CIDR Block**: `10.0.1.0/24`
   - **Availability Zone**: Choose any AZ.
3. **Private Subnet:**
   - **Name**: `private-subnet`
   - **VPC**: `prod-vpc`
   - **IPv4 CIDR Block**: `10.0.2.0/24`
4. Click **Create Subnet**.

3. Set Up Internet Gateway (IGW)

1. Navigate to **Internet Gateways** → Click **Create Internet Gateway**.
2. **Name**: `prod-igw`
3. Click **Create Internet Gateway**.
4. **Attach the IGW** to `prod-vpc` by selecting **Actions** → **Attach to VPC**.

4. Set Up NAT Gateway (NGW)

1. Navigate to **NAT Gateways** → Click **Create NAT Gateway**.
2. **Name**: `prod-ngw`
3. **Subnet**: Select `public-subnet`.
4. **Elastic IP**: Allocate a new Elastic IP.
5. Click **Create NAT Gateway**.

5. Create Route Tables

1. Navigate to **Route Tables** → Click **Create Route Table**.
2. **Public Route Table:**
   - **Name**: `prod-public-route-table`
   - **VPC**: `prod-vpc`
   - Click **Create**.
3. **Private Route Table:**

- **Name**: `prod-private-route-table`
- **VPC**: `prod-vpc`
- Click **Create**.

6. Configure Routing

1. Select **prod-public-route-table** → Click **Edit Routes**.
2. Add a new route:
   - **Destination**: `0.0.0.0/0`
   - **Target**: `prod-igw`
   - Click **Save Changes**.
3. Select **prod-private-route-table** → Click **Edit Routes**.
4. Add a new route:
   - **Destination**: `0.0.0.0/0`
   - **Target**: `prod-ngw`
   - Click **Save Changes**.

7. Associate Subnets with Route Tables

1. Select **prod-public-route-table** → Click **Subnet Associations** → Associate `public-subnet`.
2. Select **prod-private-route-table** → Click **Subnet Associations** → Associate `private-subnet`.

8. Verify Connectivity

1. Launch an EC2 instance in **public-subnet**.
2. Assign a **public IP** and verify connectivity using SSH or `ping`.
3. Launch another EC2 instance in **private-subnet**.
4. Test internet access using the NAT Gateway (`curl` or `ping` external URLs).

Completion Criteria

- A **custom VPC** named `prod-vpc` is created with designated **IPv4 CIDR block**.
- **Public and private subnets** are created and associated with appropriate **route tables**.
- **Internet Gateway (IGW) and NAT Gateway (NGW)** are set up and configured correctly.
- **Route tables** are configured to manage public and private subnet traffic.
- Successful verification of network connectivity.

Submission Guidelines

- Take **screenshots** of key steps and results.

- Compile them into a **Google Doc** and upload them to **Google Drive**.
- Share the **Google Drive link** with the instructor and ensure they have access.

🚀 **Great job! You've successfully set up a VPC with secure networking in AWS!** 🎉

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Lab

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